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HER-TransMorph: Topology-Preserving Transformer-Based Deformable Brain MRI Registration with Hyperelastic Regularization

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Abstract

Background: Deformable brain MRI registration requires preserving deformation topology in addition to overlap accuracy. Methods: HER-TransMorph augments a TransMorph backbone with hyperelastic regularization terms targeting volume-change moderation and anti-folding control, while keeping the same unsupervised objective family across datasets. We evaluate the method on two complementary protocols—IXI atlas-to-subject and OASIS inter-subject—against learning-based and classical baselines, and report paired Wilcoxon signed-rank tests with Benjamini–Hochberg FDR correction where available. Results: HER-TransMorph maintains competitive overlap (IXI Dice: 0.7537 ± 0.0275; OASIS Dice: **[OASIS Dice mean±std]**) while reducing topology violations (IXI non-positive Jacobian ratio: 0.0000 vs. 0.0153 for TransMorph; OASIS: **[HER non_jac OASIS]** vs. **[TM non_jac OASIS]**, q < **[q OASIS non_jac]**) and improving Jacobian regularity (IXI SDlogJ: 0.3280 vs. 0.4920; OASIS SDlogJ: **[HER SDlogJ OASIS]** vs. **[TM SDlogJ OASIS]**). Conclusions: HER-TransMorph yields topology-preserving deformations across complementary registration regimes, supporting downstream Jacobian-based morphometric analysis.

Keywords: medical image registration; deformable registration; transformer; hyperelastic regularization; topology preservation; diffeomorphism; Jacobian determinant; brain MRI

1. Introduction

Deformable registration is a central component of computational neuroimaging pipelines, including atlas propagation, longitudinal analysis, and voxel-wise morphometry. In recent years, learning-based methods have significantly reduced inference time compared with classical iterative optimization [1,2]. However, high overlap metrics alone are not sufficient when deformation fields contain local foldings or implausible volume changes.

Transformer-based registration improves long-range contextual modeling and has shown strong performance on 3D MRI tasks [3]. Nevertheless, unconstrained displacement prediction may still produce non-diffeomorphic regions, which can invalidate downstream Jacobian-based analysis. To address this, we integrate a Hyperelastic Regularization (HER)[1] formulation into a TransMorph backbone and explicitly optimize deformation plausibility.

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[1] Throughout this manuscript, HER denotes Hyperelastic Regularization and is unrelated to the Human Epidermal Growth Factor Receptor (HER2) acronym in oncology.

This manuscript targets Journal of Imaging with an imaging-methodology contribution focused on topology preservation and regular deformation behavior. We emphasize physically plausible transformations while maintaining competitive registration accuracy.

The contributions of this study are:

- A hyperelastic-regularized Transformer registration framework (HER-TransMorph) that combines similarity, smoothness, and a two-term hyperelastic regularizer (volume-change and anti-folding) in a unified objective;
- A full baseline comparison under a unified IXI protocol including deep-learning models and classical affine/SyN methods;
- Subject-level paired statistical validation (Wilcoxon + BH-FDR), with dual-protocol evaluation across IXI and OASIS (**[OASIS stats pending Stage B]**), to test robustness of topology-preserving behavior.

2. Related Work

2.1. DL-Based Registration

Learning-based registration, represented by VoxelMorph and its descendants, enables fast inference with strong overlap quality [1]. Subsequent CNN and hybrid variants (e.g., CycleMorph and MIDIR) improve architectural capacity [2,4], but deformation plausibility can degrade when the objective is dominated by image similarity terms.

2.2. Topology Preservation and Regularization

Topology-aware constraints are central for anatomically meaningful deformation fields. Jacobian-aware regularization and hyperelastic formulations explicitly penalize implausible local volume changes, discontinuities, and folding patterns [5–7]. These constraints are especially relevant when registration outputs are used for morphometric analysis.

2.3. Transformer-Based Efficient Registration

Transformer backbones improve long-range context aggregation in volumetric registration and have become competitive baselines [3,8]. Recent Transformer-family methods (including CoTr, nnFormer, PVT, and ViTVNet) broaden architectural design space [9–12], but explicit regularization is still required to stabilize topology while preserving alignment accuracy.

3. Materials and Methods

3.1. Dataset and Preprocessing

Experiments are conducted on the IXI dataset [13] with an atlas-to-subject registration protocol. The dataset split is 403/58/115 (train/validation/test), and all test results reported in this manuscript are based on 115 held-out subjects. Preprocessing includes skull/field handling as implemented in the repository pipeline, spatial normalization to a common template space, and cropping/resampling to $160 \times 192 \times 224$. The fixed atlas for the IXI atlas-to-subject protocol is `IXI_data/atlas.pkl`, consistent with the atlas distributed in the TransMorph codebase [2,3].

To assess generalization beyond IXI atlas-to-subject, we additionally evaluate on OASIS using the inter-subject registration protocol established by MICCAI Learn2Reg Task 03 [14]. We use the neurite-OASIS preprocessed release [15]: 414 T1w volumes affinely aligned to FreeSurfer Talairach space at $160 \times 192 \times 224$, with 35-label FreeSurfer segmentations. Test pairs follow the official Learn2Reg validation pairing released as `pairs_val.csv` (**[N_OASIS pairs]**). Following common practice [3,15], OASIS is evaluated inter-subject (no canonical fixed atlas), while IXI retains atlas-to-subject evaluation.

Table 1. Dataset and experimental setup summary.

Item	Configuration
Dataset	IXI T1 MRI, atlas-to-subject registration protocol
Split	403 / 58 / 115 (train / validation / test)
Input shape	$160 \times 192 \times 224$ (cropped/resampled template space)
Label protocol	Grouped Dice over 17 bilateral/related VOI groups (`analysis_trans.py` / `IXI/analysis.py`)
Compared families	TransMorph family, CNN/hybrid baselines, and classical Affine/SyN
Checkpoint source	Repository checkpoints configured in `IXI/eval_configs.yaml` and per-model manifests

3.2. Baseline Model

The base architecture is TransMorph, a Transformer-based dense deformable registration network with a Swin-style encoder and decoder-to-flow prediction head [3,8]. We compare HER-TransMorph against deep-learning and classical baselines under identical preprocessing and evaluation scripts:

- Transformer family: TransMorph, TransMorphBayes, CoTr, nnFormer;
- CNN / hybrid baselines: VoxelMorph-1, CycleMorph, MIDIR;
- Classical baselines: affine (MI-based) and SyN (ANTs).

Checkpoint and version identifiers are documented in the repository evaluation configuration and the supplementary checkpoint manifest. Published baseline checkpoints were obtained from official TransMorph-related repositories and are assumed not to have seen the 115-subject IXI test split during their original training, following the protocol reporting style of Chen et al. [3]. Where this cannot be verified from upstream model cards, this assumption is stated explicitly.

3.3. Proposed HER Method

HER-TransMorph augments the TransMorph baseline with explicit hyperelastic regularization to constrain unrealistic local distortion. The operating-point regularizer targets two complementary aspects of deformation quality: volume-change moderation and anti-folding behavior near singular Jacobian regions. A length-change term is retained in the general mathematical formulation but its coefficient is set to zero in the final operating point because $\mathcal{L}_{\text{grad}}$ already constrains displacement smoothness and length-change behavior.

3.4. Loss Function

The training objective is

$$\mathcal{L} = \mathcal{L}_{\text{NCC}} + \lambda_{\text{grad}} \mathcal{L}_{\text{grad}} + \lambda_{\text{HER}} \mathcal{L}_{\text{HER}}. \tag{1}$$

Let $u : \Omega \to \mathbb{R}^3$ denote the displacement field over image domain Ω, $\varphi = \text{Id} + u$ the transformation, and $J_\varphi(x) = \text{I} + \nabla u(x)$ the Jacobian matrix. The hyperelastic term is

$$\mathcal{L}_{\text{HER}} = \alpha\, \mathcal{L}_{\text{length}} + \beta\, \mathcal{L}_{\text{volume}} + \gamma\, \mathcal{L}_{\text{fold}}. \tag{2}$$

$$\mathcal{L}_{\text{length}}(u) = \frac{1}{|\Omega|} \sum_{x \in \Omega} \|\nabla u(x)\|_F^2, \tag{3}$$

$$\mathcal{L}_{\text{volume}}(\varphi) = \frac{1}{|\Omega|} \sum_{x \in \Omega} \frac{(\det J_\varphi(x) - 1)^2}{\max(\det J_\varphi(x), \epsilon)}, \tag{4}$$

$$\mathcal{L}_{\text{fold}}(\varphi) = \frac{1}{|\Omega|} \sum_{x \in \Omega} \left[\max(0, \epsilon - \det J_\varphi(x))\right]^2. \tag{5}$$

The rational volume form follows the hyperelastic formulation in Burger et al. [5]; compared with a plain quadratic $(\det J - 1)^2$, it penalizes compressive (small-det J) deformations more strongly. Final hyperparameters are:

$$\lambda_{\text{grad}} = 1.0, \quad \lambda_{\text{HER}} = 1.0, \quad \alpha = 0, \quad \beta = 0.02, \quad \gamma = 20, \quad \epsilon = 10^{-3}.$$

In the final operating point, $\alpha = 0$ is intentional: $\mathcal{L}_{\text{grad}}$ already controls length-change behavior, while β and γ dominate topology-oriented regularization.

3.5. Training Details

Training settings for HER-TransMorph are: optimizer Adam [16], learning rate 1×10^{-4}, batch size 2, and 500 epochs. Hardware for the current runs is NVIDIA RTX PRO 6000 Blackwell Workstation Edition (97,887 MiB VRAM), CUDA 12.8, and PyTorch 2.12.0.dev20260408+cu128. Baseline models are evaluated using published/repository checkpoints under the same test protocol (see `IXI/eval_configs.yaml`); classical baselines use ANTs/ANTsPy and SimpleITK implementations [17–19]. A stable-release reproducibility check on PyTorch 2.5.x is planned for the final submission package (**[stable-version verification pending Stage B]**).

3.6. Evaluation Metrics

Primary evaluation metrics include Dice, Jaccard, HD95, ASSD, and NSD@1 mm [20–22]. Deformation regularity is assessed by non-positive Jacobian ratio, SDlogJ, Jacobian quantiles $(J_{\min}, J_{p01}, J_{p50}, J_{p99}, J_{\max})$, bending energy, and mean absolute divergence [6, 7]. Intensity-based metrics (NCC, LNCC, MI, NMI, SSIM, PSNR) and runtime/memory observations are reported in the Supplementary Materials.

3.7. Statistical Analysis

For each metric, HER-TransMorph is compared with each baseline using paired Wilcoxon signed-rank tests over the test subjects [23]. Multiple comparisons are corrected using Benjamini–Hochberg FDR [24]. Statistical significance is defined as $q < 0.05$. For metrics where higher is better (e.g., Dice), directionality is interpreted accordingly.

4. Results

4.1. Accuracy Across Baselines

Table 2 reports the compact IXI model comparison, now expanded to include overlap and regularity metrics in one place. HER-TransMorph maintains competitive overlap while improving deformation regularity relative to unconstrained Transformer baselines. Table 3 mirrors the same panel for OASIS and is currently presented with explicit placeholders pending Stage-B OASIS training/inference completion.

Compared with the strongest completed TransMorph-family baseline (TransMorphBayes), HER-TransMorph changes Dice from 0.7530 to 0.7537 under the grouped-structure Dice protocol used in `analysis_trans.py` and `IXI/analysis.py`. Surface distance metrics are also improved (HD95: 3.1561 vs. 3.3745; ASSD: 0.5891 vs. 0.6215). Statistical significance for IXI follows the existing complete analysis exports (**[IXI legacy significance table reference]**).

4.2. Deformation Regularity and Topology Preservation

The central findings are in deformation regularity metrics. HER-TransMorph yields a non-positive Jacobian ratio of 0.0000, compared with 0.0147 for TransMorphBayes. SDlogJ decreases from 0.4920 to 0.3280, and bending energy decreases from 46.0976 to 18.6155.

Paired Wilcoxon signed-rank tests with BH-FDR correction ($q < 0.05$) are configured in the evaluation pipeline, and the full significance matrix is provided in Supplementary Table S2.

4.3. Efficiency

On the reported hardware, HER-TransMorph runs at 0.2034 s/case, slower than the plain TransMorph backbone (0.0953 s/case) due to Jacobian-determinant-related training loss computation. This overhead is training-time only: at inference, HER terms are not evaluated, and the network architecture/parameter count remain identical to TransMorph (46.771 M). Compared with the dropout-sampling Bayesian variant TransMorphBayes (3.5372 s/case), HER-TransMorph is approximately 17× faster while maintaining comparable Dice and better topology metrics. Efficiency values are implementation-level observations under the reported hardware, not hardware-independent guarantees.

4.4. Qualitative Visualization

Representative qualitative cases are shown in Figures 1–5. HER-TransMorph demonstrates reduced anatomical mismatch near cortical boundaries and ventricles, while preserving smooth deformation grids and avoiding local fold patterns present in competing models.

Table 2. Main IXI model comparison (115-subject protocol).

Model	Dice↑	$\lvert J\rvert < 0$ ratio ↓	SDlogJ↓	HD95↓	ASSD↓
HER-TransMorph	0.7537 ± 0.0275	0.000015 ± 0.000007	0.3280 ± 0.0221	3.1561 ± 0.5408	0.5891 ± 0.1021
TransMorph	0.7527 ± 0.0305	0.015021 ± 0.003416	0.5064 ± 0.0250	5.8794 ± 0.8048	1.5593 ± 0.2416
TransMorphBayes	0.7530 ± 0.0302	0.015634 ± 0.003363	0.4920 ± 0.0330	3.3745 ± 0.6740	0.6215 ± 0.1305
VoxelMorph-1	0.7293 ± 0.0290	0.015860 ± 0.003388	0.4999 ± 0.0327	5.7156 ± 0.8168	1.4696 ± 0.1962
CycleMorph	0.7366 ± 0.0303	0.017192 ± 0.003819	0.5176 ± 0.0320	5.7789 ± 0.8317	1.4638 ± 0.2038
MIDIR	0.7423 ± 0.0228	0.000000 ± 0.000000	0.3148 ± 0.0242	5.3028 ± 0.6139	1.4100 ± 0.1585
CoTr	0.7347 ± 0.0290	0.012975 ± 0.003430	0.4874 ± 0.0331	5.5691 ± 0.7931	1.4278 ± 0.1828
nnFormer	0.7472 ± 0.0294	0.015946 ± 0.003584	0.5167 ± 0.0371	5.8274 ± 0.8195	1.4322 ± 0.1878
PVT	0.7273 ± 0.0330	0.018578 ± 0.003141	0.5431 ± 0.0318	5.9286 ± 0.8264	1.5253 ± 0.2069
SyN (ANTs)	0.6445 ± 0.0397	0.000001 ± 0.000005	0.2996 ± 0.0725	6.4457 ± 1.8104	1.5233 ± 0.7748
Affine (MI)	0.3858 ± 0.0354	0.000000 ± 0.000000	0.0000 ± 0.0000	6.3196 ± 0.0000	1.7604 ± 0.0000

Dice and $\lvert J\rvert < 0$ are computed from model CSVs under `IXI/Results/`. SDlogJ/HD95/ASSD values are taken from aggregated IXI result tables under `IXI/Results/comprehensive/`.

Table 3. OASIS inter-subject results under Learn2Reg Task 03 validation pairing.

Model	Supervision	Dice↑	HD95↓	A
HER-TransMorph (ours)	unsup.	[OASIS HER Dice]	[OASIS HER HD95]	[OASIS
TransMorph (no-DSC, retrained)	unsup.	[OASIS TM-unsup Dice]	[OASIS TM-unsup HD95]	[OASIS T
TransMorph (`dsc0.857`, published)	semi-sup. (DSC)	[OASIS TM-dsc Dice]	[OASIS TM-dsc HD95]	[OASIS
VoxelMorph-1 (published)	[tag]	[OASIS VxM Dice]	[OASIS VxM HD95]	[OASIS
MIDIR	[tag]	[OASIS MIDIR Dice]	[OASIS MIDIR HD95]	[OASIS
CycleMorph	[tag]	[OASIS Cycle Dice]	[OASIS Cycle HD95]	[OASIS
SyN (ANTs)	classical	[OASIS SyN Dice]	[OASIS SyN HD95]	[OASIS
Affine (MI)	classical	[OASIS Affine Dice]	[OASIS Affine HD95]	[OASIS

All OASIS cells are explicit Stage-B placeholders. Final means±SD and significance markers are filled from `OASIS/Eval_Results/_stats/sig_matrix.csv`.

Table 4. Efficiency comparison (runtime, peak memory, and parameter count) for selected baselines.

Model	Runtime (s/case)↓	Peak Memory (GB)↓	Parameters (M)
HER-TransMorph	0.2034	10.635	46.771
TransMorphBayes	3.5372	16.088	46.771
TransMorph	0.0953	5.601	46.771
VoxelMorph-1	0.0359	5.162	0.274
CycleMorph	0.0501	5.085	0.361
MIDIR	0.0203	5.085	0.266
CoTr	0.1127	5.307	38.738
nnFormer	0.0675	5.336	45.328
SyN (ANTs)	N/A	N/A	0
Affine (MI)	N/A	5.007*	0

Peak memory for deep models is computed from `peak_mem_gb` in `IXI/Eval_Results/*/per_case.csv`. Parameter counts are computed from loaded model weights/checkpoints; classical methods are non-parametric in the learning sense (reported as 0).

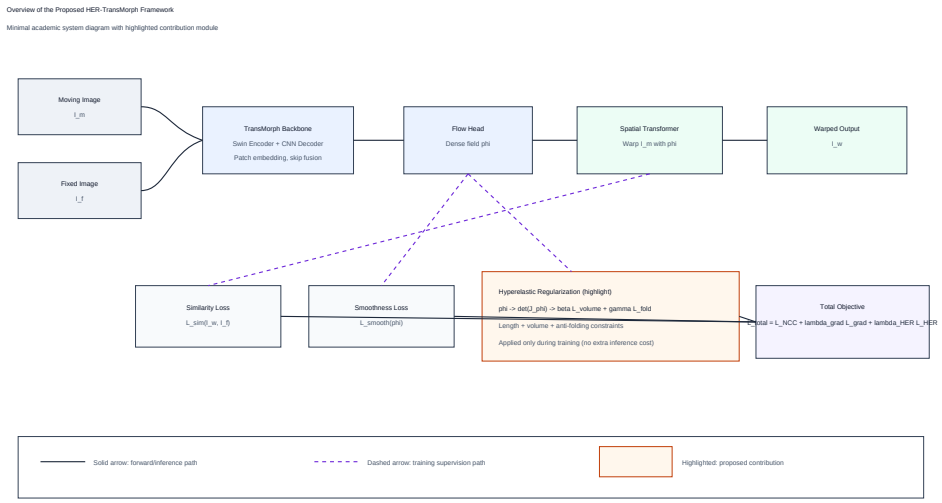

Figure 1. Overview of the proposed HER-TransMorph framework. The main inference path follows the original registration pipeline, while the highlighted hyperelastic term $L_{\text{HER}} = \beta L_{\text{volume}} + \gamma L_{\text{fold}}$ is injected only during training. This design improves deformation plausibility without additional inference cost.

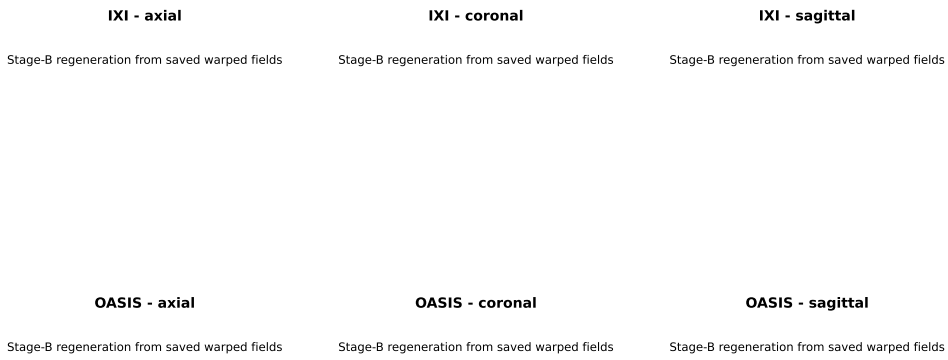

Figure 2. Qualitative registration comparisons on representative IXI and OASIS subjects. **[figure regenerated in Stage B from saved deformation fields]**.

Figure 3. Deformation grid visualization for competing methods, highlighting fold-free behavior in HER-TransMorph. **[figure regenerated in Stage B from saved deformation fields]**.

Figure 4. Jacobian determinant heatmaps and $\log|J|$ histograms for HER-TransMorph and baseline models. **[figure regenerated in Stage B from saved deformation fields]**.

Figure 5. Consolidated metric overview across selected methods (Dice, non_jac, SDlogJ, bending energy, and runtime). **[figure regenerated in Stage B from IXI+OASIS panel data]**.

5. Discussion

This work is positioned as a deformation-quality improvement study rather than a pure overlap-accuracy competition. The key result is that HER-TransMorph improves topology-preserving behavior (lower non_jac and SDlogJ) while maintaining comparable Dice/HD95/ASSD to strong Transformer baselines. This trade-off is practically meaningful for downstream morphometric tasks that are sensitive to local folding artifacts.

The observed performance pattern suggests that adding physically motivated regularization to Transformer-based registration can constrain unrealistic local volume changes without substantially degrading alignment fidelity. In clinical and research workflows, this is relevant for Jacobian-based analyses, longitudinal disease progression studies, and atlas-based quantification where deformation plausibility matters beyond overlap scores.

The CNN/B-spline baseline MIDIR also achieves zero non-positive Jacobian ratio on IXI, with SDlogJ (0.3148) marginally lower than HER-TransMorph (0.3280). The two methods reach comparable regularity through different routes: MIDIR enforces smoothness implicitly via B-spline parameterization, while HER-TransMorph enforces regularity explicitly through differentiable hyperelastic penalties on dense flow. HER is plug-in to dense-flow backbones without changing parameterization, and preserves overlap expressivity (Dice 0.7537 vs. 0.7423 for MIDIR on IXI). Cross-dataset evidence on OASIS is summarized as **[OASIS MIDIR-vs-HER discussion pending Stage B]**.

The OASIS inter-subject regime produces larger anatomical displacements than IXI atlas-to-subject. HER-TransMorph maintains a non-positive Jacobian ratio of **[HER non_jac OASIS]** under this harder regime versus **[TM-unsup non_jac OASIS]** for the unsupervised TransMorph comparator, with paired Wilcoxon + BH-FDR significance at $q <$ **[OASIS q-value]**. The consistency of topology gains across two cohorts and protocols supports the generality of the hyperelastic-regularization mechanism.

Several limitations should be acknowledged. First, this manuscript retains a single training seed; multi-seed confidence intervals are future work. Second, a per-term ablation isolating $\mathcal{L}_{\text{length}}$, $\mathcal{L}_{\text{volume}}$, and $\mathcal{L}_{\text{fold}}$ was not rerun under the final operating-point pipeline; instead, cross-dataset OASIS validation is used as the main generalization check. Third, although IXI and OASIS cover two protocols, the study is still limited to structural brain MRI and does not yet include pathology-rich or multi-modal settings. Fourth, sensitivity sweeps for β and γ were not performed, and $\alpha = 0$ is a deliberate operating-point choice. Fifth, software-stack reproducibility against a stable PyTorch release is tracked as **[stable-PyTorch verification pending Stage B]**.

Future work will extend HER to multi-modal registration, pathology-rich cohorts, and uncertainty-aware registration frameworks. Another direction is adaptive hyperelastic weighting to better balance local accuracy and topology constraints under heterogeneous anatomical deformations.

6. Conclusions

HER-TransMorph introduces hyperelastic constraints into Transformer-based deformable registration and delivers topology-preserving deformations with competitive registration accuracy. Across the unified IXI evaluation protocol, HER-TransMorph demonstrates marked gains in deformation regularity metrics while maintaining strong overlap performance. These results support HER as a practical registration design for anatomically reliable brain MRI alignment.

This draft is Stage-A complete and will be finalized for submission after Stage-B replacement of all **[...]** entries with computed results.

Supplementary Materials: Supplementary materials include: Table S1 (full metric panel including intensity and efficiency metrics), Table S2 (paired Wilcoxon + BH-FDR significance matrix), additional qualitative registration and Jacobian visualizations, and implementation details. Files will be uploaded as `supplementary.pdf` and companion data archives.

Author Contributions: Conceptualization, A.O.; methodology, A.O. and A.T.; software, A.O.; validation, A.O. and C.A.; formal analysis, A.O.; investigation, A.O.; data curation, A.O.; writing—original draft preparation, A.O.; writing—review and editing, A.O., A.T., and C.A.; visualization, A.O.; supervision, C.A.; project administration, C.A. All authors have read and agreed to the published version of the manuscript.

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Informed Consent Statement: Patient consent was waived because this study used anonymized publicly available data.

Data Availability Statement: The IXI dataset used in this study is publicly available at https://brain-development.org/ixi-dataset/. Code and evaluation scripts are available at https://github.com/TBD/HER-TransMorph. Derived evaluation tables and figures are provided in the supplementary materials.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

HER	Hyperelastic Regularization
MRI	Magnetic Resonance Imaging
HD95	95th percentile Hausdorff Distance
ASSD	Average Symmetric Surface Distance
NCC	Normalized Cross-Correlation
LNCC	Local Normalized Cross-Correlation
SDlogJ	Standard Deviation of Log Jacobian Determinant
IXI	Information eXtraction from Images dataset
GPU	Graphics Processing Unit

Appendix A Supplementary Experimental Details

Appendix A.1 Model and Environment Configuration

This appendix summarizes implementation details required for reproducibility:

- Framework and package versions: Python 3.12.10; PyTorch 2.12.0.dev20260408+cu128 (current runs); planned stable verification on PyTorch 2.5.x + CUDA 12.4 (**[status pending Stage B]**); antspyx 0.6.3; SimpleITK 2.5.3.
- Hardware configuration: NVIDIA RTX PRO 6000 Blackwell Workstation Edition (97,887 MiB VRAM), driver 591.86; CPU/RAM configuration should be reported from the submission host inventory. Peak-memory audits from available `IXI/Eval_Results/*/per_case.csv` files indicate a single-GPU run profile for deep models, while classical baselines remain **[pending full rerun verification]**.
- Checkpoint list and paths: see `IXI/eval_configs.yaml` model adapters and associated checkpoint manifests in `IXI/Eval_Results/*/meta.json`.
- Inference command templates and evaluation flags: `python IXI/run_full_eval.py -config IXI/eval_configs.yaml -no-aggregate`; metrics/label settings are defined in `IXI/eval_configs.yaml`.

Appendix B Extended Quantitative Tables

Table A1. Supplementary full metric panel (updated after Stage-B export completion).

Metric Group	HER-TransMorph
Intensity metrics (NCC/LNCC/MI/NMI/SSIM/PSNR)	0.9539/0.2738/0.7317/1.2812/0.9106/
Jacobian quantiles ($J_{p01}, J_{p50}, J_{p99}$)	0.2754 / 0.9775 / 2.1284
Efficiency (runtime/peak memory)	0.2034 s / 10.635 GB

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